CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2012 series

5054 PHYSICS

5054/41 Paper 4 (Alternative to Practical), maximum raw mark 30

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1 **(a)** **(i)** horizontal level or point marked level with top of hook B1 [1]

(ii) eye / E labelled level with dotted line B1 [1]

(b) within extension of spring /
within elastic limit / not permanently stretched /
gives smooth oscillations /
load does not jumps off spring / spring does not become slack B1 [1]

(c) reduces human reaction error (in *T)* / more accurate *T* /
T too small /
gives average value (of *T*) B1 [1]

(d) 8.024 / 8.02 / 8.0 seen OR $\Sigma t \div 10$ C1
0.4012 / 0.401 / 0.40 (s) A1 [2]

(e) **(i)** 0.401 written in table ecf (**d**) (3 sf required) B1 [1]

(ii) axes: correct way round, labelled quantity and unit

scales: linear, not awkward, more than ½ grid B1
e.g. *x*-axis: 2 cm ≡ 1 N *y*-axis: 2 cm ≡ 0.1 s

points plotted accurately within ½ small square
neat crosses or small points (in circle) B1

smooth curve of best fit neatly drawn B1 [4]

(iii) yes + when $W = 0$ there will be (no extension so) no oscillations B1 [1]

(allow no + when $W = 0$ there will be some extension due to mass of spring)

(iv) non-linear with *T* increasing as *W* increases B1 [1]

[Total: 13]

2 **(a)** **(i)** lamp lights (normal brightness) B1 [1]

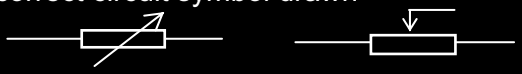
(ii) any one from:
broken wire / connections not good
lamp blown / faulty
cell(s) run down B1 [1]

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(b) (i) lamp becomes dimmer B1 [1]

(ii) **1.** rheostat / variable resistor / potentiometer B1 [1]

2. correct circuit symbol drawn B1 [1]



3. wire is coiled B1 [1]

[Total: 6]

3 **(a)** withstand (high) pressure / force (from air) (outside) B1 [1]

(b) seals bell-jar / prevents air entering B1 [1]

(c) (i) sound gradually becomes quieter B1
sound cannot travel through a vacuum / requires medium / air B1 [2]

(ii) light can travel through a vacuum / does not require medium / air B1 [1]

(d) sound / vibrations can travel through the metal plate B1 [1]

[Total: 6]

4 **(a)** to determine height accurately / to stop as soon as shoe moves B1 [1]

(b) 22° ± 1° B1 [1]

(c) (i) any one sensible suggestion, e.g.
protractor has edge
protractor is small
divisions close together
alignment of zero difficult
board sags
board may move B1 [1]

(ii) **measures** two sides of triangle and uses trig formula B1 [1]
(may be shown on diagram)

(d) (better grip) larger angle / ramp lifted higher or reverse argument B1 [1]

[Total: 5]

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